

# Jun Seok Kim

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[github.com/forestinblue](https://github.com/forestinblue)

## EDUCATION

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### The Chinese University of Hong Kong, Shenzhen

Sep 2024 – Expected Mar 2027

Master of Science in Artificial Intelligence and Robotics

GPA: 3.11/4.0 (coursework in progress)

Main Courses: Image Processing and Computer Vision · Deep Learning · Capstone Project (Grade: Distinction)

### Kangwon National University

Mar 2017 – Feb 2023

Bachelor of Science, Information Statistics

Bachelor of Science in Engineering, Convergence Program of Data Science (Double Major)

GPA: 3.63/4.5 (90.1%)

Main Courses: Machine Learning · Theory of Probability · Multivariate Statistical Analysis

**Research Interests:** Robust perception for robotics, uncertainty-aware 3D perception

## PUBLICATIONS & PREPRINTS

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[1] **J. Kim** and R. Huang. Range-aware uncertainty calibration in self-supervised monocular depth.

In preparation for CVPR 2027

## RESEARCH EXPERIENCE

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**MSc Student Researcher**, Computer Vision Laboratory (Advisor: Prof. [Rui Huang](#))

Sep 2024 – Present

**Range-aware uncertainty calibration in self-supervised monocular depth (MSc thesis).**

Jan 2026 – Current

- **Self-supervised depth models** never see ground-truth depth, so their predicted uncertainty is never tied to actual error. On KITTI, standard estimators under-report error at every range, most severely at near range where **robots need it most**, even MonoProb (WACV 2024), which was designed for interpretable uncertainty.
- **Calibrated** it with **no ground-truth depth at all**: trained only the uncertainty head on residuals against classical stereo matching, then set its scale with one label-free constant.
- Cut near-range **calibration mismatch** by 83% versus the original model, and by 59% versus rescaling with LiDAR ground truth.

**LocalAggTR: Differentiable Voxelization for Weakly Supervised 3D Occupancy**

Sep 2025 – Dec 2025

[\[code\]](#)

- GaussTR's Gaussian-to-voxel step is non-differentiable, so 3D occupancy losses cannot train the network end-to-end, and it dominates inference time.
- Replaced it with a differentiable **Local Aggregation** operator that passes occupancy gradients.
- **7.2× faster inference** with mIoU unchanged on nuScenes. **Adopted by the lab's 3D Gaussian Splatting project** for semi-supervised fine-tuning.

**Geometric vs. Generative 3D Augmentation for Point Cloud Classification**

Jan 2025 – May 2025

[\[code\]](#)

- Tested whether text-to-3D generative augmentation helps a point cloud classifier more than geometric augmentation.
- Built the Point-E generation pipeline and a geometric augmentation module, then ran a configuration ablation with PointNet on ModelNet40.
- **Geometric augmentation** raised accuracy by 6.5%, while text-to-3D augmentation lowered it by 1.5% and collapsed performance when used alone.

## Point Cloud Density and 3D Object Detection

Jun 2025– Jul 2025

### [\[code\]](#)

- Asked how a **3D object detector** degrades when LiDAR density drops and whether upsampling recovers it.
- Built a resampling pipeline that LiDAR points by random or voxel-grid downsampling and restores them with two upsampling methods, then evaluated PointPillars on nuScenes-mini.
- **Voxel-grid downsampling** kept one fifth of the points with only a 2.1% relative mAP drop, nearly four times smaller than random downsampling to half. Upsampling recovered less than half of the random-sampling loss.

## INDUSTRY EXPERIENCE

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### Founder, [TARS](#)

Feb 2024 – Present, South Korea

- Built and operate a **YOLO-based foot-traffic analytics** service for small businesses such as restaurants, cafés and real-estate offices, measuring hourly pass-by, stop and entry rates from storefront cameras.
- Deployed to +60 paying clients alongside my studies, turning the measurements into customer-segment consulting reports; self-funded and profitable since launch.

### Backend & Infrastructure Engineer, [Bluzen Corp.](#)

Jul 2022 – Aug 2022, Singapore

- Designed a MariaDB database for construction workforce records and built its Node.js backend.
- Supported on-site build-out of Naver server rooms at the Singtel Data Center, handling cabling and network setup.

### Data Scientist, [Connple Corp.](#)

Jun 2021 – Jun 2022, South Korea

- Built and shipped a **BERT-based restaurant recommender** to a **production mobile app**, paired with SVD collaborative filtering on review ratings.
- Extracted food keywords from Instagram reviews with LDA topic modeling to power a food discovery service.
- Automated nationwide collection of 600,000 restaurant records through a scheduled Flask data pipeline.

## SCHOLARSHIPS AND HONORS

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- **Mid-Sized Company Employment Support Scholarship** (full tuition + living stipend, 3 semesters), Korea Student Aid Foundation (KOSAF) — 2021–2023
- **Passion Award**, 21st & 22nd Gangwon Province University Student Entrepreneurship Competition – 2020, 2021
- **Commendations for exemplary service and field construction**, Republic of Korea Army — 2019

## EXTRACURRICULAR ACTIVITIES

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### President, Photo Club, Kangwon National University

Jan 2021 – Jun 2021

- Led a 100+ member club through COVID-19, teaching photography fundamentals and splitting large outings into small-group formats to keep activities running.

### Team Lead, Student Startup Teams, Kangwon National University

Jan 2020 – Jun 2021

- Led teams that built prototypes of an AI travel planner and a restaurant AI recommendation app, winning the Passion Award at the Gangwon Province University Student Entrepreneurship Competition two years in a row.

### Squad Leader, 930th Artillery Battalion, Republic of Korea Army

Apr 2018 – Nov 2019

- Led a 10-person squad in artillery operations and field construction (mandatory military service).

## SKILLS & LANGUAGES

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- Programming: Python, C/C++, CUDA, SQL
- Frameworks: PyTorch, MMDetection3D, MMEngine, scikit-learn
- Tools: Git, Linux, Docker
- Languages: Korean (native), English (full professional), Mandarin Chinese (basic)